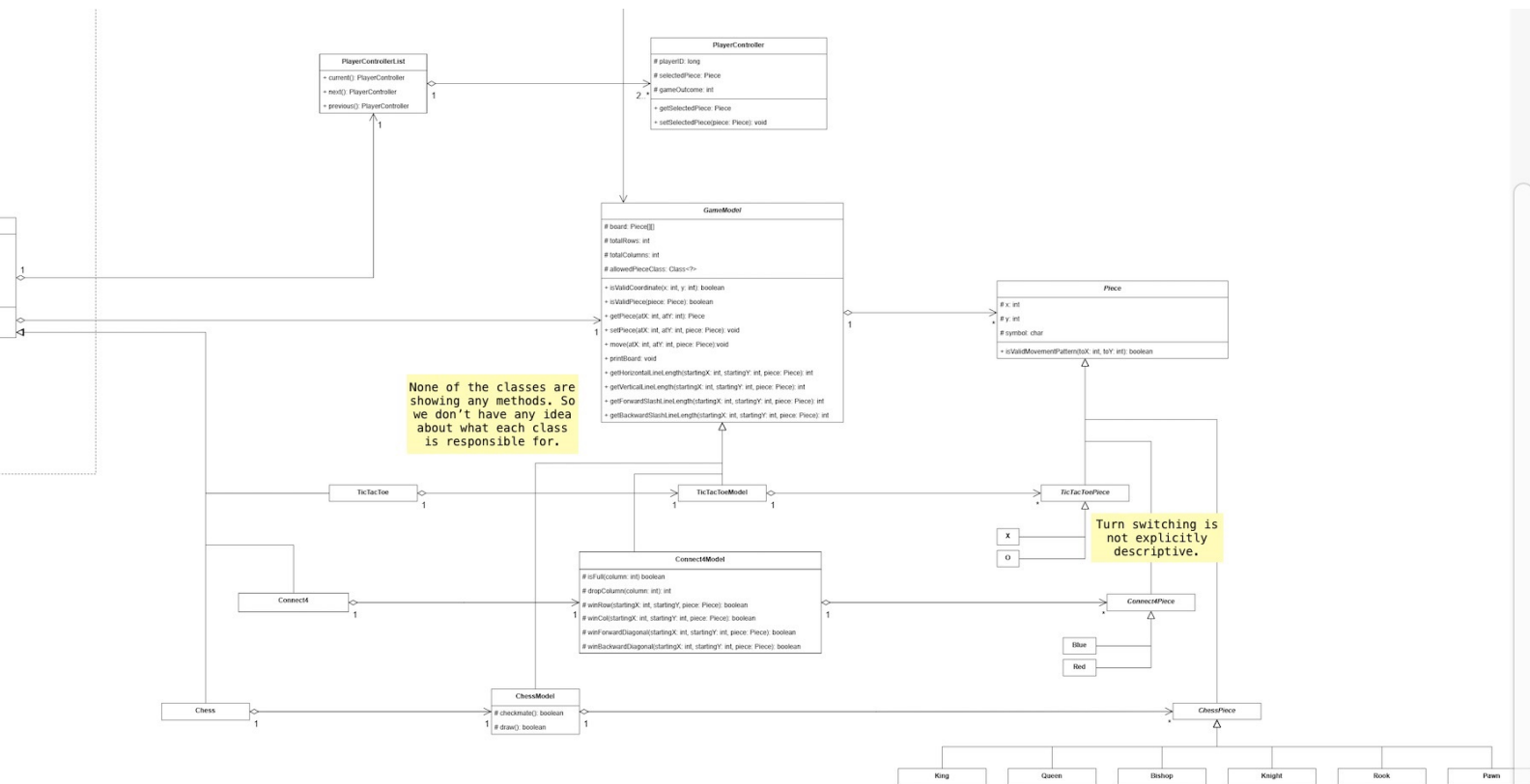
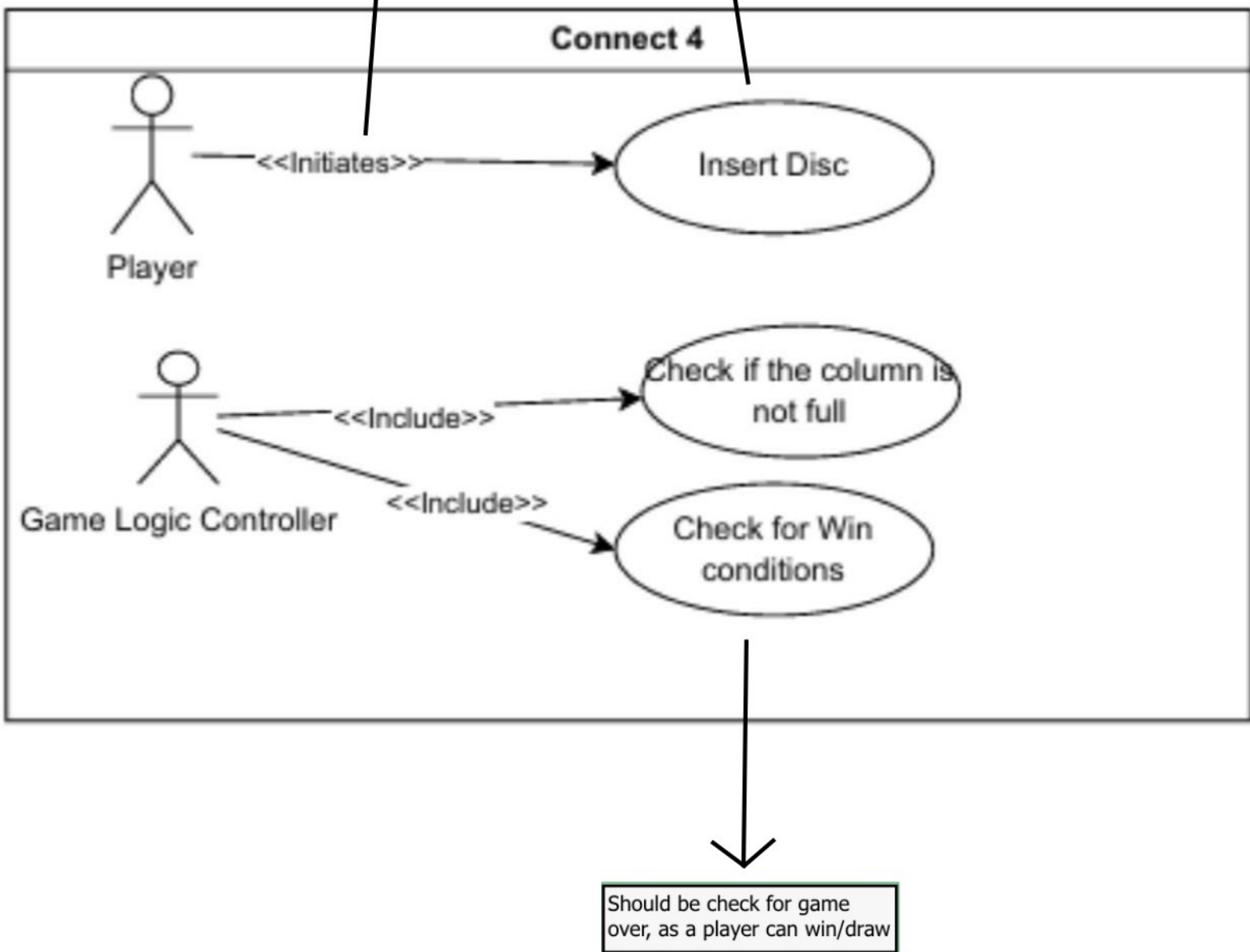
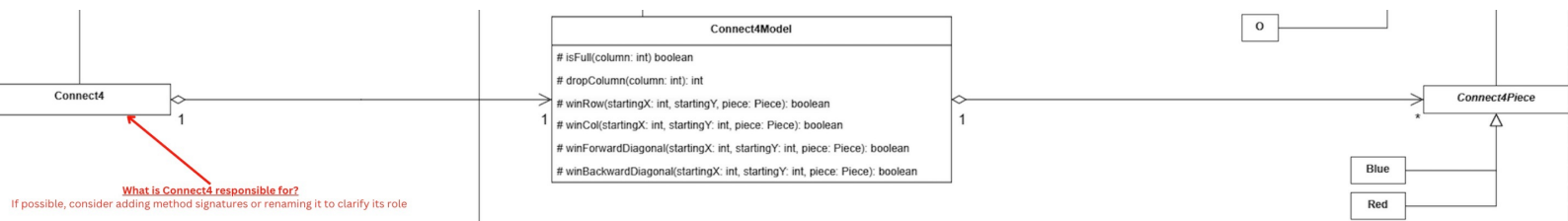




The label "<<initiates>>" is not standard UML

Insert disc and check if column should be connected





# Networking Use Cases

## Session Timeout

Iteration: 1, last modification: March 07, 2025

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description      | Start a timer that count down the player's inactive time and automatically                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Primary Actor    | Player                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Goal in Context  | Ensure inactive players are automatically disconnected to free up server resources and maintain a smooth multiplayer experience.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Preconditions    | <ol style="list-style-type: none"><li>1. The player is connected to the server and in an active game session.</li><li>2. A session timeout policy is defined (e.g., disconnect after 5 minutes of inactivity).</li></ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Trigger          | The player has been inactive for a predefined period.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Scenario         | <div>The system does not distinguish between a player intentionally quitting vs. accidental disconnections due to internet loss or misclicks. Suggestion: could allow players a window of time to reconnect even through manual accidental disconnection.</div> <ol style="list-style-type: none"><li>1. The player stops interacting with the game (no moves, chat, or other actions).</li><li>2. The server starts a countdown for the session timeout.</li><li>3. The server sends a warning message to the player about an impending connection.</li><li>4. If no response is received, the server disconnects the player from the game.</li><li>5. The server updates the game state and notifies the remaining players.</li></ol> |
| Exceptions       | <ol style="list-style-type: none"><li>1. If the player resumes activity before the timeout, the countdown is reset.</li><li>2. If the disconnection was due to network issues, an automatic reconnection attempt is made.</li></ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Priority         | High (ensures smooth server performance and fair gameplay).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Frequency of Use | Occasionally (depends on player inactivity)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Channel to Actor | <div>Inconsistent formatting, fix formatting for clarity</div> <ol style="list-style-type: none"><li>1. : Manages the session timeout and disconnection process.</li></ol> <div>Other Players: Receive a notification that the player has been removed</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                         |                                                                                                                                                                                                    |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Secondary Actors</b> | GameServerManager, Other players                                                                                                                                                                   |
| <b>Open Issues</b>      | <ol style="list-style-type: none"><li>1. Should disconnected players be allowed to rejoin the same session?</li><li>1. How long should the server wait before removing inactive players?</li></ol> |

Inconsistent  
formatting, fix  
formatting for clarity



# Network Connection

Iteration: 1, last modification: March 7, 2025

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Primary Actor</b>           | Player                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Goal in Context</b>         | Ensure the player can establish and maintain a stable connection with the game server for real-time interactions.                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Preconditions</b>           | <ul style="list-style-type: none"><li>• The player has a stable internet connection.</li><li>• The game client is launched and ready to connect</li></ul>                                                                                                                                                                                                                                                                                                                                                      |
| <b>Trigger</b>                 | The player starts the game and attempts to connect to the server.                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Scenario</b>                | <p>The player opens the game and selects "Connect to Server." The game client sends a connection request to the server.</p> <p>The server authenticates the request and establishes a session. If successful, the server sends a confirmation message.</p> <p>The player is now connected and can proceed to matchmaking or gameplay.</p>                                                                                                                                                                      |
| <b>Exceptions</b>              | <ul style="list-style-type: none"><li>• If the connection request fails:<ul style="list-style-type: none"><li>◦ The player receives an error message (e.g., "Server Unreachable")</li><li>◦ The system may retry the connection automatically</li></ul></li><li>• If the player loses connection mid-game:<ul style="list-style-type: none"><li>◦ The system attempts reconnection within a set time.</li><li>◦ If unsuccessful, the player is marked as disconnected in the game session.</li></ul></li></ul> |
| Could better define 'set time' |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Priority</b>                | Critical (network stability is essential for multiplayer functionality).                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Frequency of Use</b>        | Every time a player starts a session.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Channel to Actor</b>        | In-game UI notifications (connection status, error messages).                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Secondary Actors</b>        | GameServerManager, Authentication System.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Open Issues</b>             | <ul style="list-style-type: none"><li>• How many reconnection attempts should be allowed before disconnecting the player?</li></ul>                                                                                                                                                                                                                                                                                                                                                                            |

# Update States

Iteration: 1, last modification: March 7, 2025

Error in formatting, fix formatting for clarity

|                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                   |
| <b>Primary Actor</b>                                                                                                                                                                                                 | Player                                                                                                                                                                                                                                                            |
| <b>Goal in Context</b>                                                                                                                                                                                               | <div>6. Update the states of the game synchronously</div> <div>7. Enhance the user experience with seamless transitions/ changes</div> <div>8. Handle delays accordingly and minimize delays</div>                                                                |
| <b>Preconditions</b>                                                                                                                                                                                                 | <div>1. The game is running and the network is established</div> <div>2. The player has a stable internet connection</div>                                                                                                                                        |
| <b>Trigger</b>                                                                                                                                                                                                       | <div>1. The player makes a move in the game</div> <div>2. The players interact with each other in the game</div>                                                                                                                                                  |
| <b>Scenario 1</b>                                                                                                                                                                                                    | <div>1. The player makes a valid move(change) in the game.</div> <div>2. This triggers the “Update states” use case.</div> <div>3. The game server updates the changes in the interface of the other players.</div>                                               |
| <b>Scenario 2</b> <div>It would be better to use different update functions for game and chat updates as in this case, the entire game state would update for every chat message, causing immense server load.</div> | <div>1. The players send messages to each other through in-game chat.</div> <div>2. This triggers the “Update states” use case.</div> <div>3. The game server updates the messages in the interface of all players through web socket servers.</div>              |
| <b>Exceptions</b> <div>Missing rollback mechanism for failed updates. If an update fails mid-process, how does the system recover? Suggestion: Could return to the previous game state if</div>                      | <div>1. If the connection of a player fails, it will cause some delays in updating the states, and the player will potentially leave the server temporarily</div> <div>2. If the server is packed, the server may face some issues with updating the states</div> |
| <b>Priority</b>                                                                                                                                                                                                      | Critical. This feature is the most fundamental aspect of a multiplayer game                                                                                                                                                                                       |
| <b>Frequency of Use</b>                                                                                                                                                                                              | Multiple Times                                                                                                                                                                                                                                                    |

# Verify Data

Iteration: 1, last modification: March 7, 2025

Major formatting error, cannot see data



|                  |                                                                                                                                                                                                                                                                                                               |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description      |                                                                                                                                                                                                                                                                                                               |
| Primary Actor    | Player                                                                                                                                                                                                                                                                                                        |
| Goal in Context  | Authorize the log-in process to make sure that only the intended user can access the account                                                                                                                                                                                                                  |
| Preconditions    | <div><div>1. The game is running and the network is established</div><div>2. The database is running properly</div><div>3. The player has a stable internet connection</div></div>                                                                                                                            |
| Trigger          | <div><div>1. The player clicks ‘Log-in’ button to access the account</div></div>                                                                                                                                                                                                                              |
| Scenario         | <div><div>1. The player tries to log into the account by providing necessary personal information</div><div>2. The system retrieves the data from the database corresponding to the provided information</div><div>3. The system compares the two data and decides to authorize the player or not</div></div> |
| Exceptions       | <div><div>1. The log-in attempt is rejected if the information does not match</div><div>2. If there is no existing data regarding the player, the player can sign up for a new account</div><div>3. The log-in process will fail if the database is corrupted</div></div>                                     |
| Priority         | Critical. This feature is the most fundamental aspect of a multiplayer game                                                                                                                                                                                                                                   |
| Frequency of Use | Multiple times                                                                                                                                                                                                                                                                                                |
| Channel to Actor | GUI, Game server                                                                                                                                                                                                                                                                                              |
| Secondary Actors | WebSocket server                                                                                                                                                                                                                                                                                              |
| Open Issues      | <div><div>1. What security measures should be added to the system to ensure the integrity of the data?</div><div>2. What action should be taken if the player fails 5 times to log in?</div><div>3. What account recovery methods should be in the system?</div></div>                                        |

# Update Data

Iteration: 1, last modification: March 7, 2025

|                                                             |                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description                                                 |                                                                                                                                                                                                                                                                                                                                                |
| Primary Actor                                               | Player                                                                                                                                                                                                                                                                                                                                         |
| Goal in Context                                             | Update the data in the database                                                                                                                                                                                                                                                                                                                |
| Preconditions                                               | <ol style="list-style-type: none"><li>1. The game is running and the network is established</li><li>2. The database is running properly</li><li>3. The player has a stable internet connection</li><li>4. The player is already logged into the account successfully</li></ol>                                                                 |
| Trigger                                                     | <ol style="list-style-type: none"><li>1. The player clicks 'Update' button to update the user information</li><li>2. The player clicks 'Save' button to save the current states of the game</li><li>3.  <span>Formatting error, empty space</span></li></ol> |
| Scenario 1                                                  | <ol style="list-style-type: none"><li>1. The player makes some changes to the user information</li><li>2. The player clicks 'Update' button to update the information</li><li>3. The system updates the user information in the database</li></ol>                                                                                             |
| Scenario 2                                                  | <ol style="list-style-type: none"><li>1. The player wants to save the current state of the game to resume at where it left off</li><li>2. The player clicks 'Save' button to save the states</li><li>3. The system updates the saved states in the database</li></ol>                                                                          |
| Exceptions<br><div>How would you handle update fails?</div> | <ol style="list-style-type: none"><li>1. No update is done if the player's information has no changes.</li><li>2. Update is rejected if the information is in an unacceptable format</li><li>3. If the database fails to connect, update cannot be done</li></ol>                                                                              |
| Priority                                                    | Critical. This feature is the most fundamental aspect of a multiplayer game                                                                                                                                                                                                                                                                    |
| Frequency of Use                                            | Multiple times                                                                                                                                                                                                                                                                                                                                 |
| Channel to Actor                                            | GUI, Game server                                                                                                                                                                                                                                                                                                                               |
| Secondary Actors                                            | WebSocket server                                                                                                                                                                                                                                                                                                                               |

# Retrieve Data

Iteration: 1, last modification: March 7, 2025

|                         |                                                                                                                                                                                                                                                                                                                        |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>      |                                                                                                                                                                                                                                                                                                                        |
| <b>Primary Actor</b>    | Player                                                                                                                                                                                                                                                                                                                 |
| <b>Goal in Context</b>  | Retrieve the data from the database for various purposes                                                                                                                                                                                                                                                               |
| <b>Preconditions</b>    | <ol style="list-style-type: none"><li>1. The game is running and the network is established</li><li>2. The database is running properly</li><li>3. The player has a stable internet connection</li></ol> <ol style="list-style-type: none"><li>1. The player is already logged into the account successfully</li></ol> |
| <b>Trigger</b>          | The player make actions which requires the system to retrieve data from the database                                                                                                                                                                                                                                   |
| <b>Scenario 1</b>       | <ol style="list-style-type: none"><li>1. The player logs into the account</li><li>2. The player provides the log-in credentials</li><li>3. The system retrieves data from the database to verify</li></ol>                                                                                                             |
| <b>Scenario 2</b>       | <ol style="list-style-type: none"><li>1. The player opens the leaderboard</li><li>2. The system retrieves data regarding the leaderboard from the database</li></ol>                                                                                                                                                   |
| <b>Scenario 3</b>       | <ol style="list-style-type: none"><li>1. The player starts a game and wait for other players to join</li><li>2. The system accesses the database to find the players whose ranks match with the player who is in the game lobby</li></ol>                                                                              |
| <b>Exceptions</b>       | <ol style="list-style-type: none"><li>1. The requested data does not exist in the database</li><li>2. The user lacks the necessary permissions to retrieve the data</li><li>3. The request contains incorrect or malformed filters</li></ol>                                                                           |
| <b>Priority</b>         | Critical. This feature is the most fundamental aspect of a multiplayer game                                                                                                                                                                                                                                            |
| <b>Frequency of Use</b> | Multiple times                                                                                                                                                                                                                                                                                                         |
| <b>Channel to Actor</b> | GUI, Game server                                                                                                                                                                                                                                                                                                       |
| <b>Secondary Actors</b> | WebSocket server                                                                                                                                                                                                                                                                                                       |

Consider adding an exception for handling database timeouts or connection failures to prevent data retrieval failures from crashing the system.

# In-Game Chat System

Iteration: 1, last modification: March 7, 2025

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description      |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Primary Actor    | Player                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Goal in Context  | Enable real-time communication between players in a game session.<br>Ensure chat messages are delivered efficiently and securely.                                                                                                                                                                                                                                                                                                                       |
| Preconditions    | <div>1. Players are connected to the game server</div> <div>2. The game session is active</div> <div>3. The chat system is enabled</div>                                                                                                                                                                                                                                                                                                                |
| Trigger          | A player types and sends a message in the in-game chat                                                                                                                                                                                                                                                                                                                                                                                                  |
| Scenario         | <div>1. The player enters a message in the chat box</div> <div>2. The client sends the message to the GameServerManager using WebSockets.</div> <div>3. The GameServerManager processes and filters the message (e.g., checks for inappropriate content).</div> <div>4. The server broadcasts the validated message to all players in the session.</div> <div>5. Each player's client receives the message and displays it in the chat window. 6.</div> |
| Exceptions       | <div>1. Network failure: The message is delayed or not sent.</div> <div>2. Server overload: Chat messages experience a delay.</div> <div>2. Inappropriate content detected: The message is blocked or censored.</div>                                                                                                                                                                                                                                   |
| Priority         | High (important for player communication and engagement)                                                                                                                                                                                                                                                                                                                                                                                                |
| Frequency of Use | Multiple times per game session.                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Channel to Actor | In-game chat UI.                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Secondary Actors | GameServerManager, Other players                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Open Issues      | <div>1. Should chat history be stored after the game ends?</div> <div>2. Should players be able to mute other players?</div> <div>3. How should spam messages be handled?</div>                                                                                                                                                                                                                                                                         |

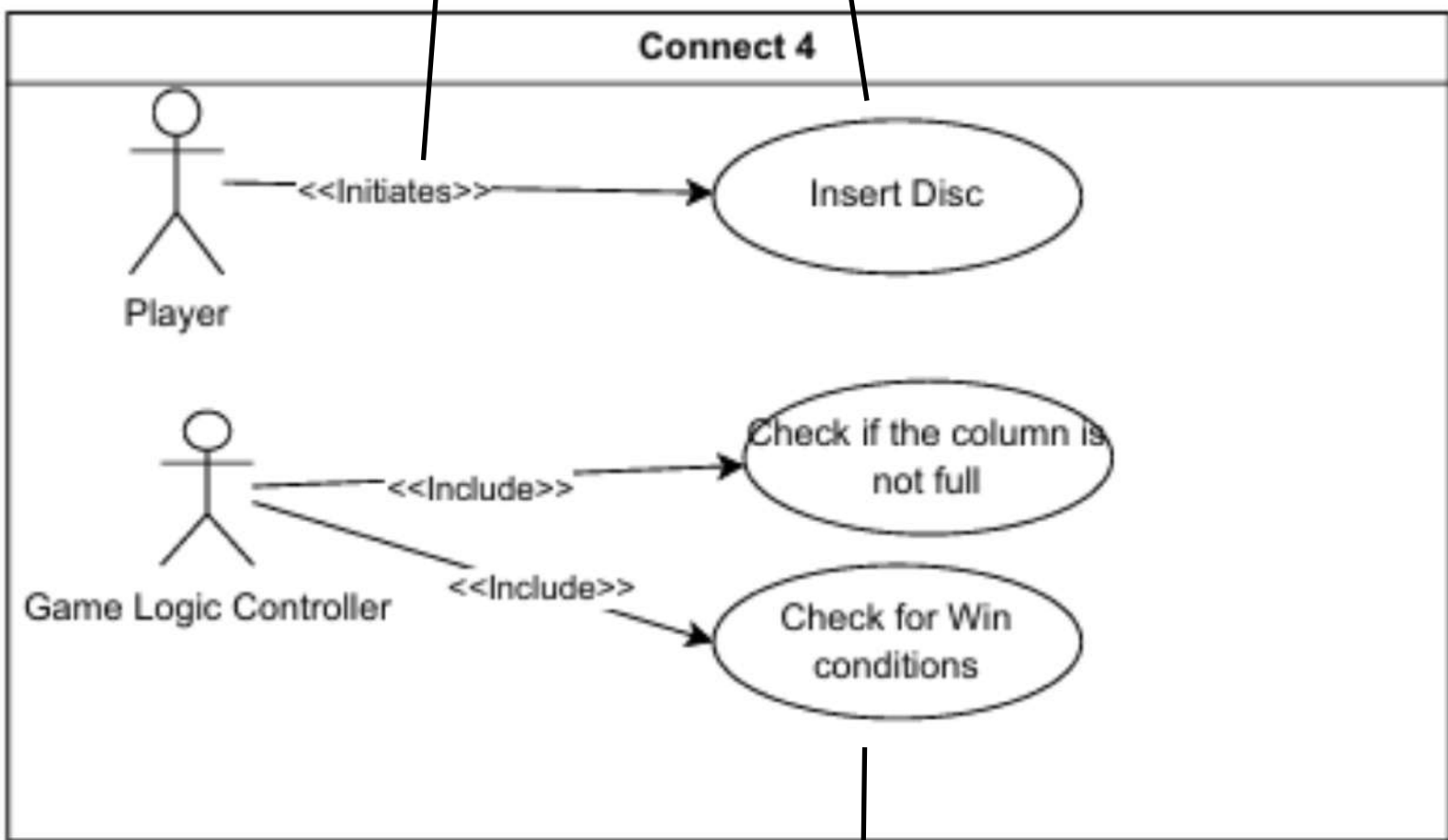
Consider adding a spam detection mechanism to prevent players from flooding the chat with messages

Inconsistent formatting

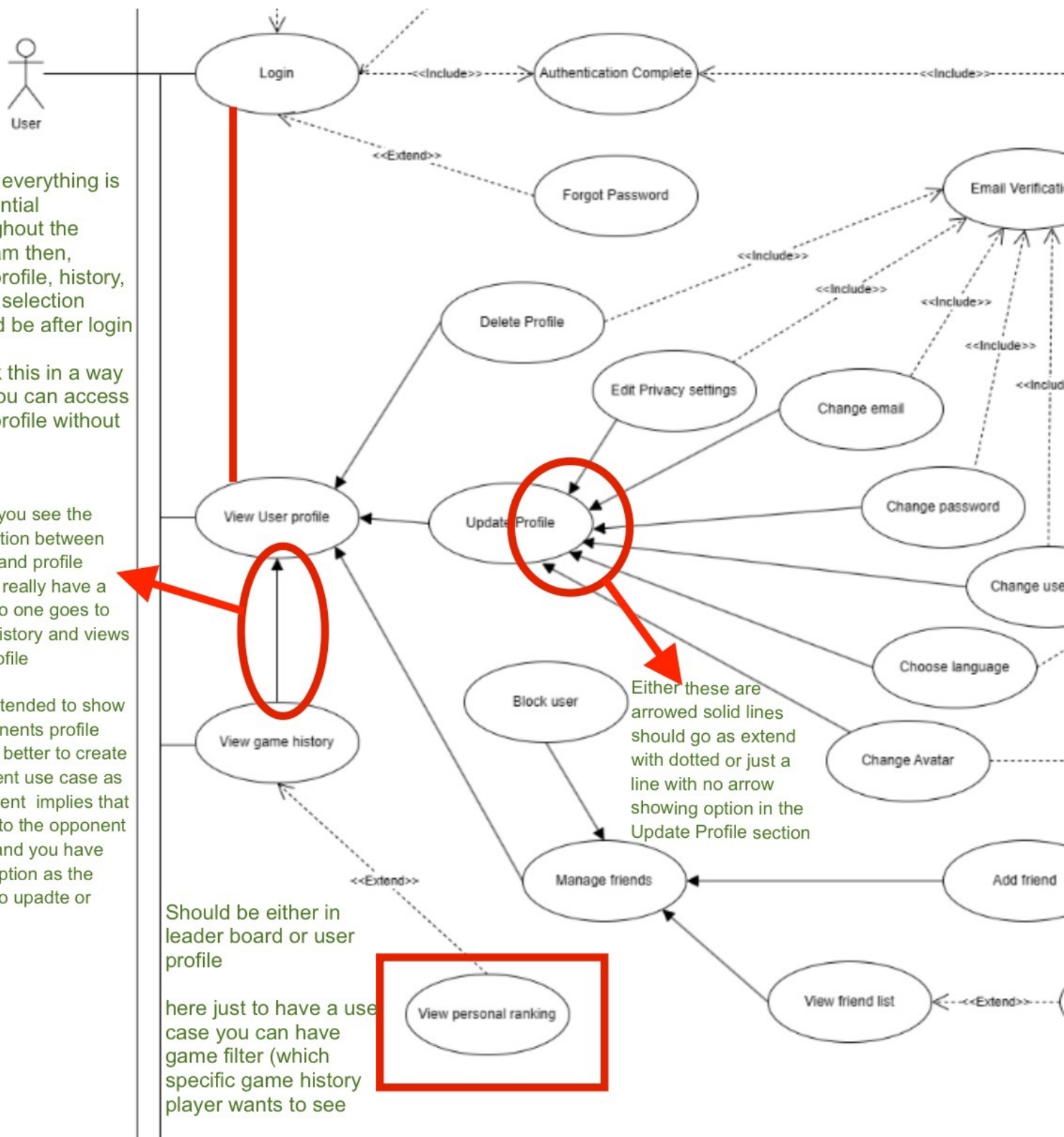
If a player disconnects and reconnects, should they receive missed messages? Define whether chat history is temporarily stored for reconnecting users.

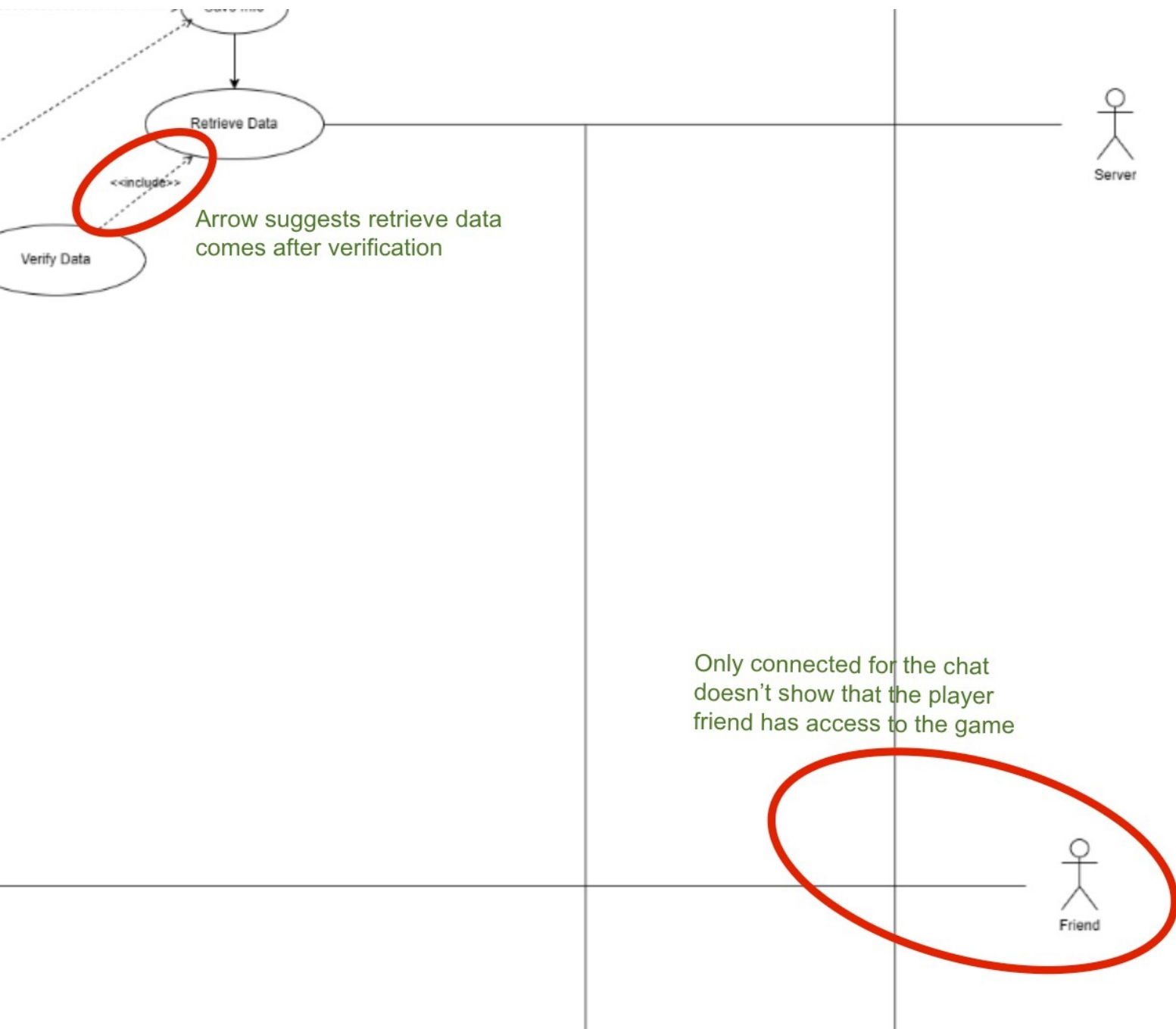
The label "<<initiates>>" is not standard UML

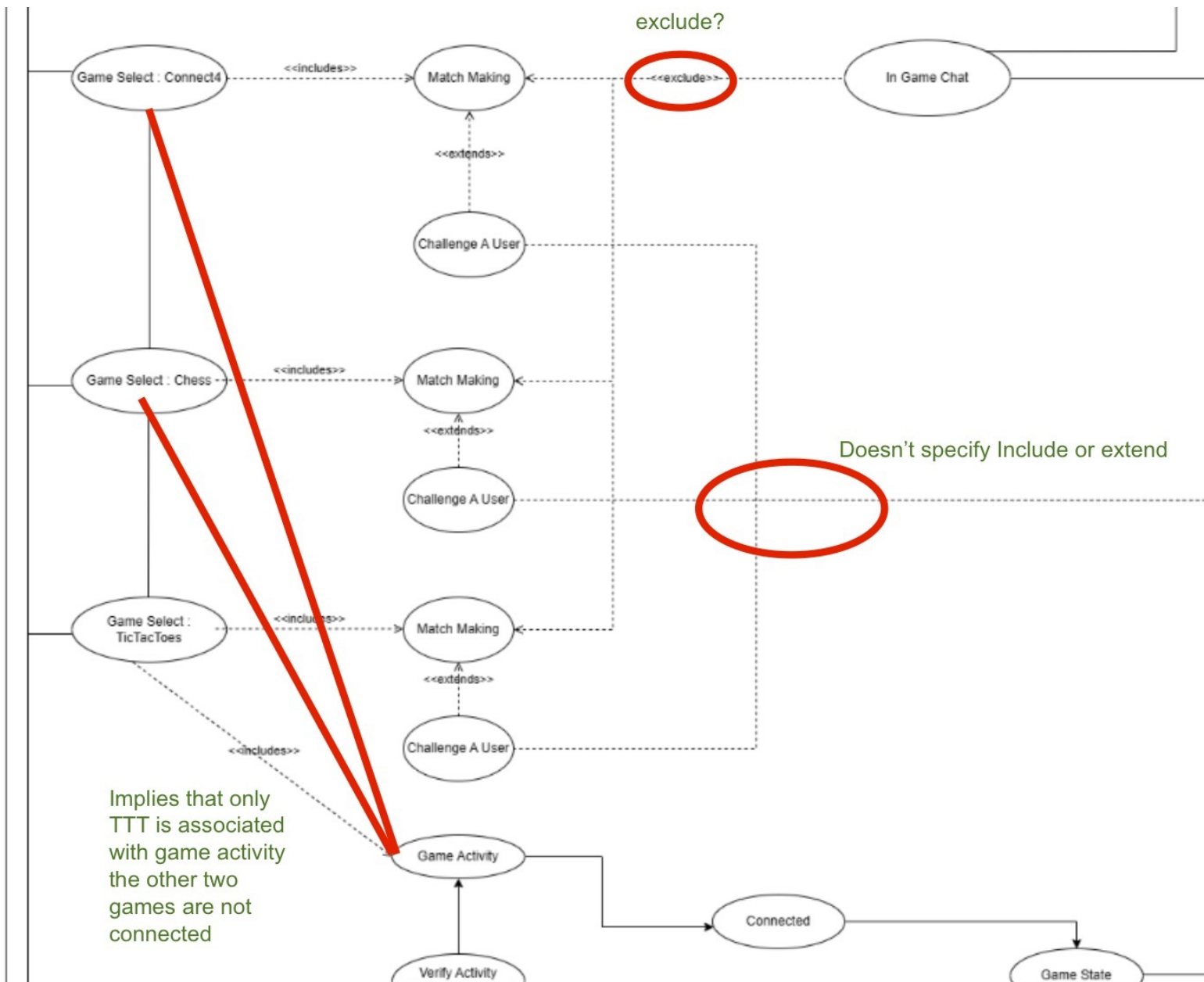
Insert disc and check if column should be connected

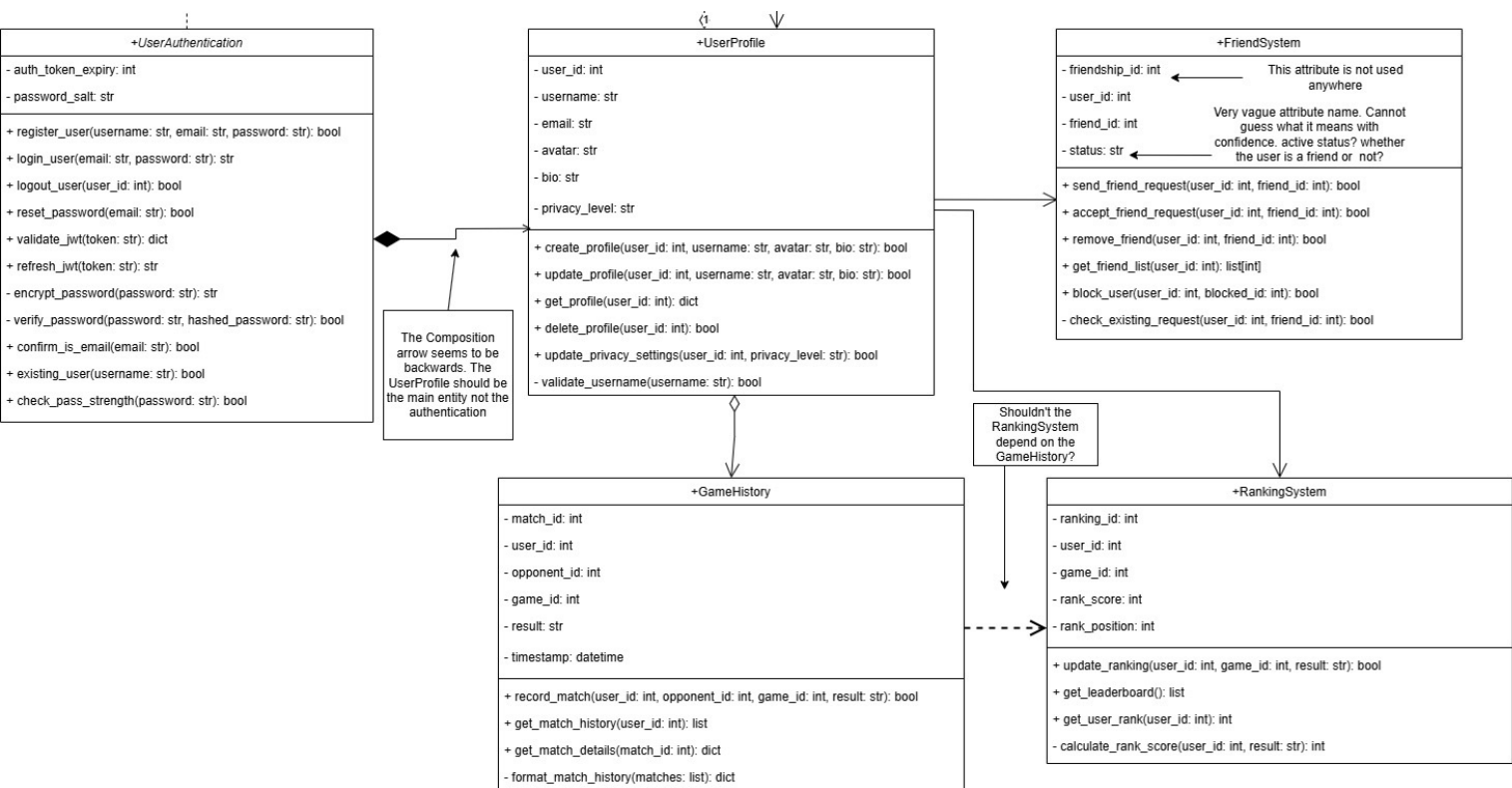


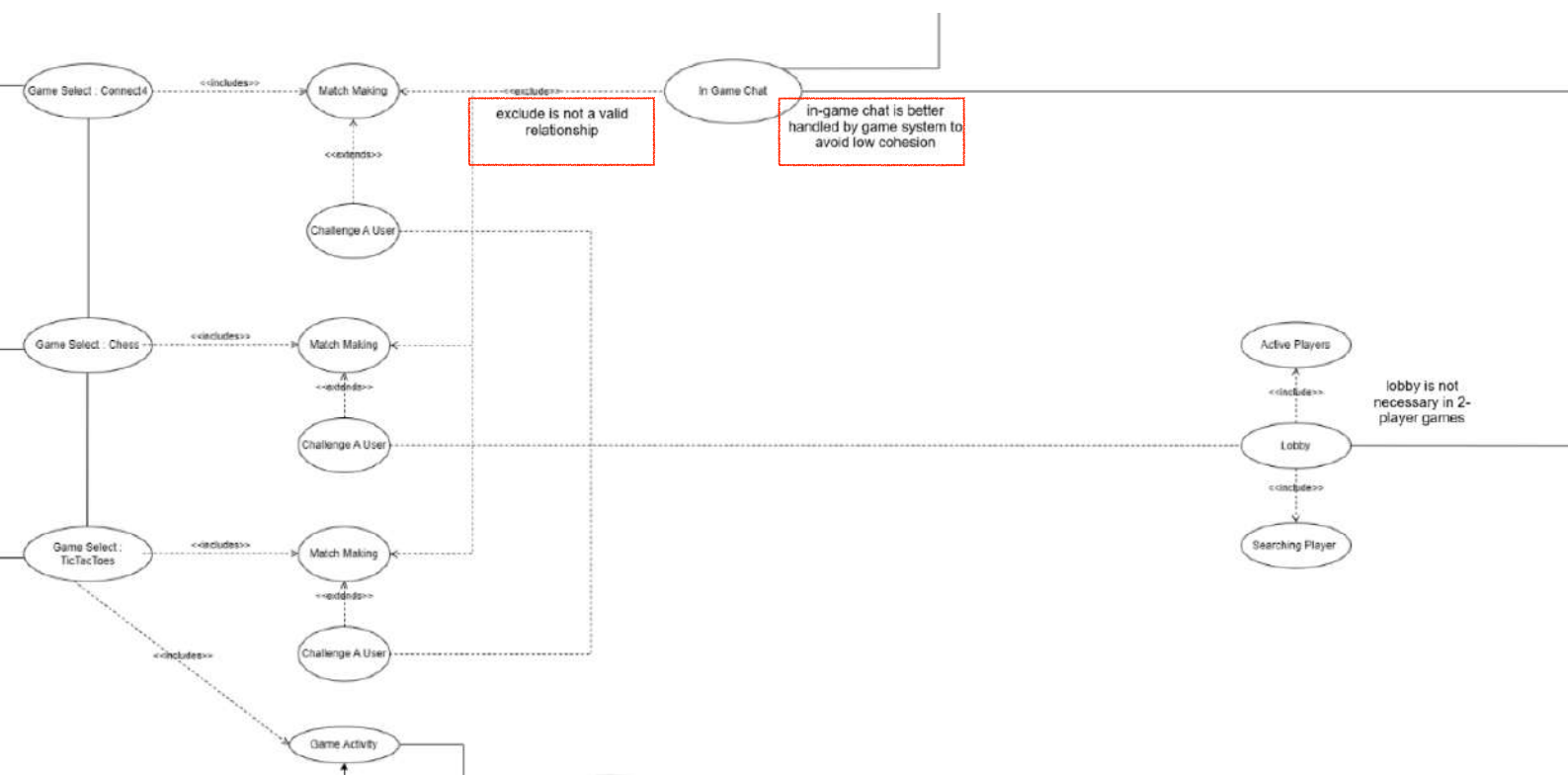
Should be check for game over, as a player can win/draw



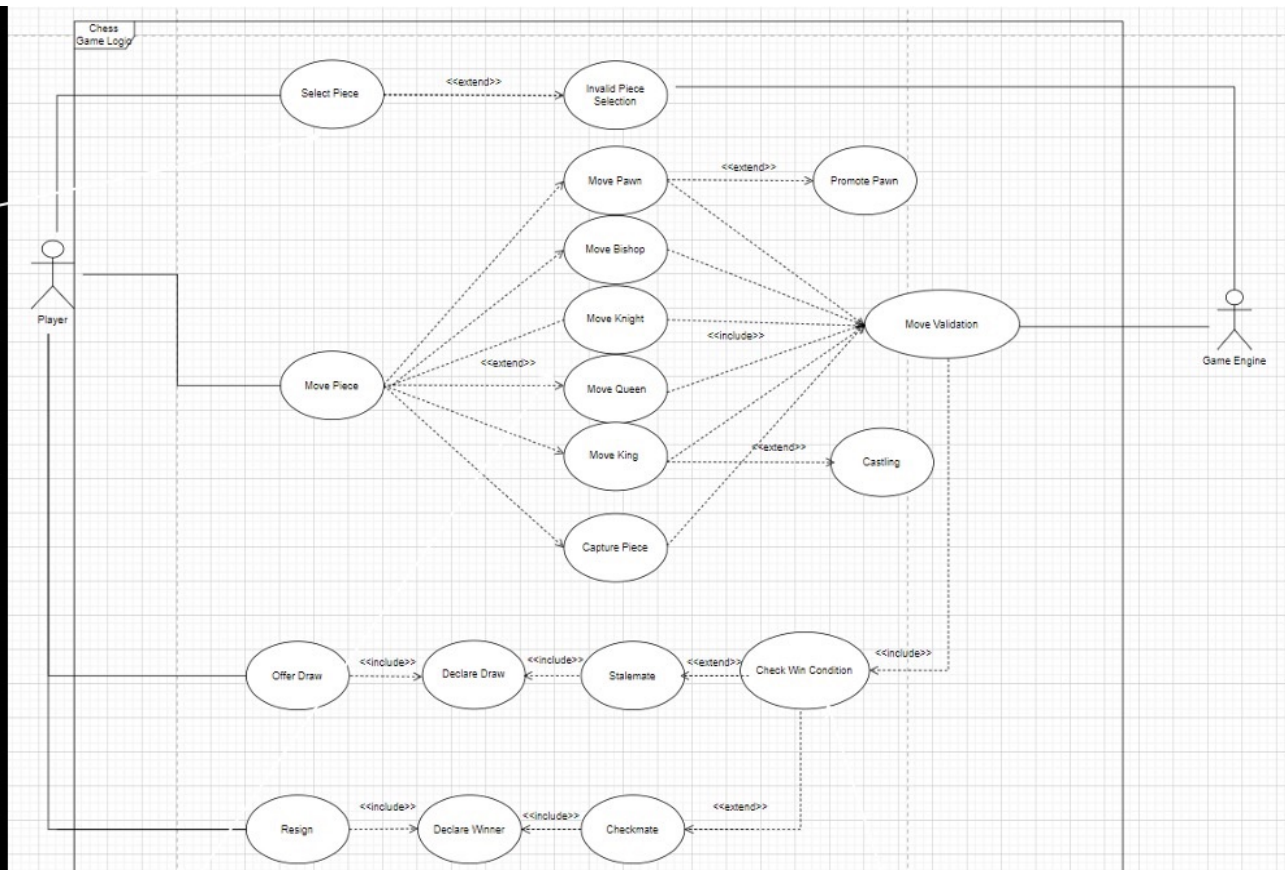








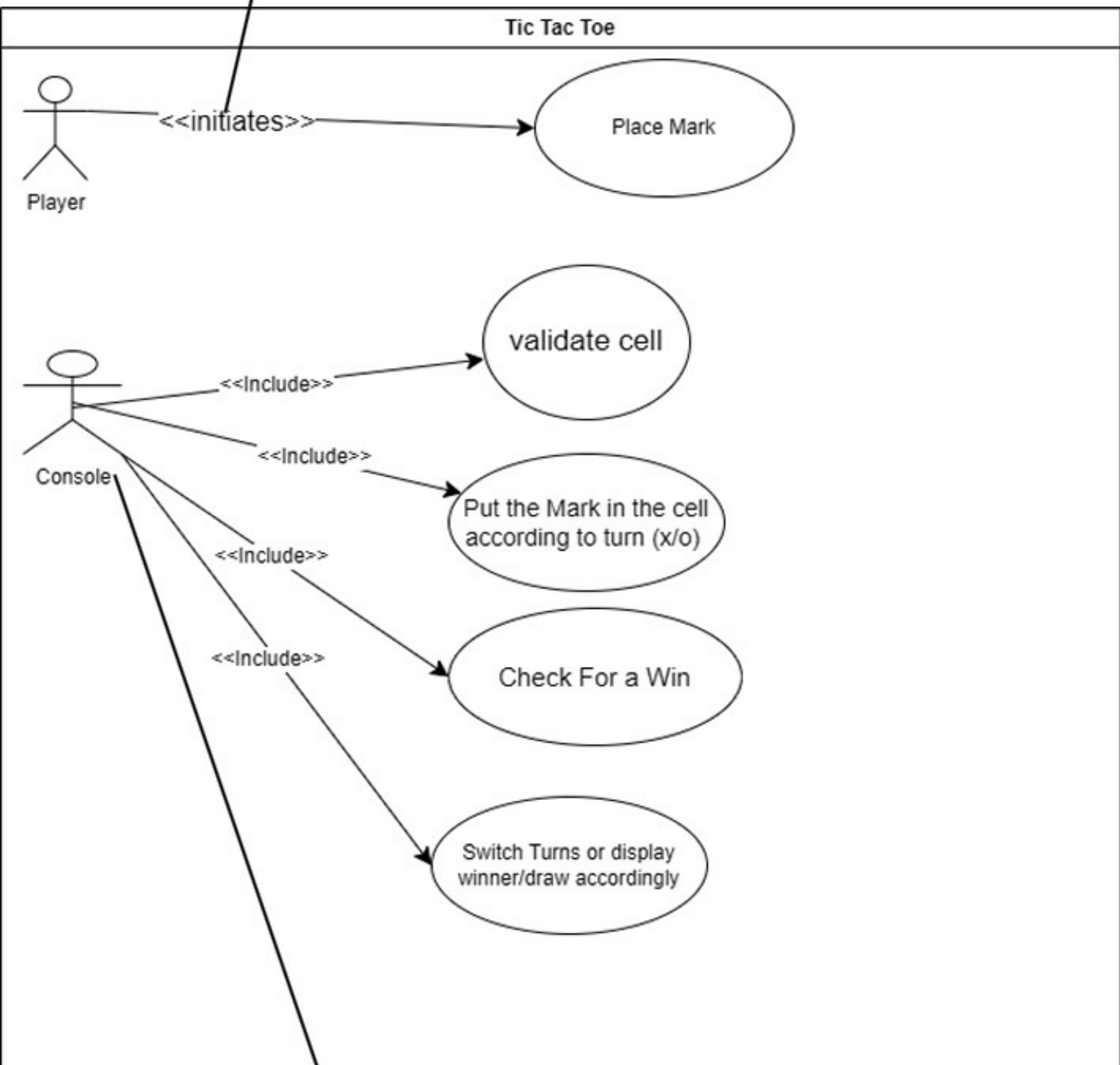
The select piece use case should also extend the move piece use case



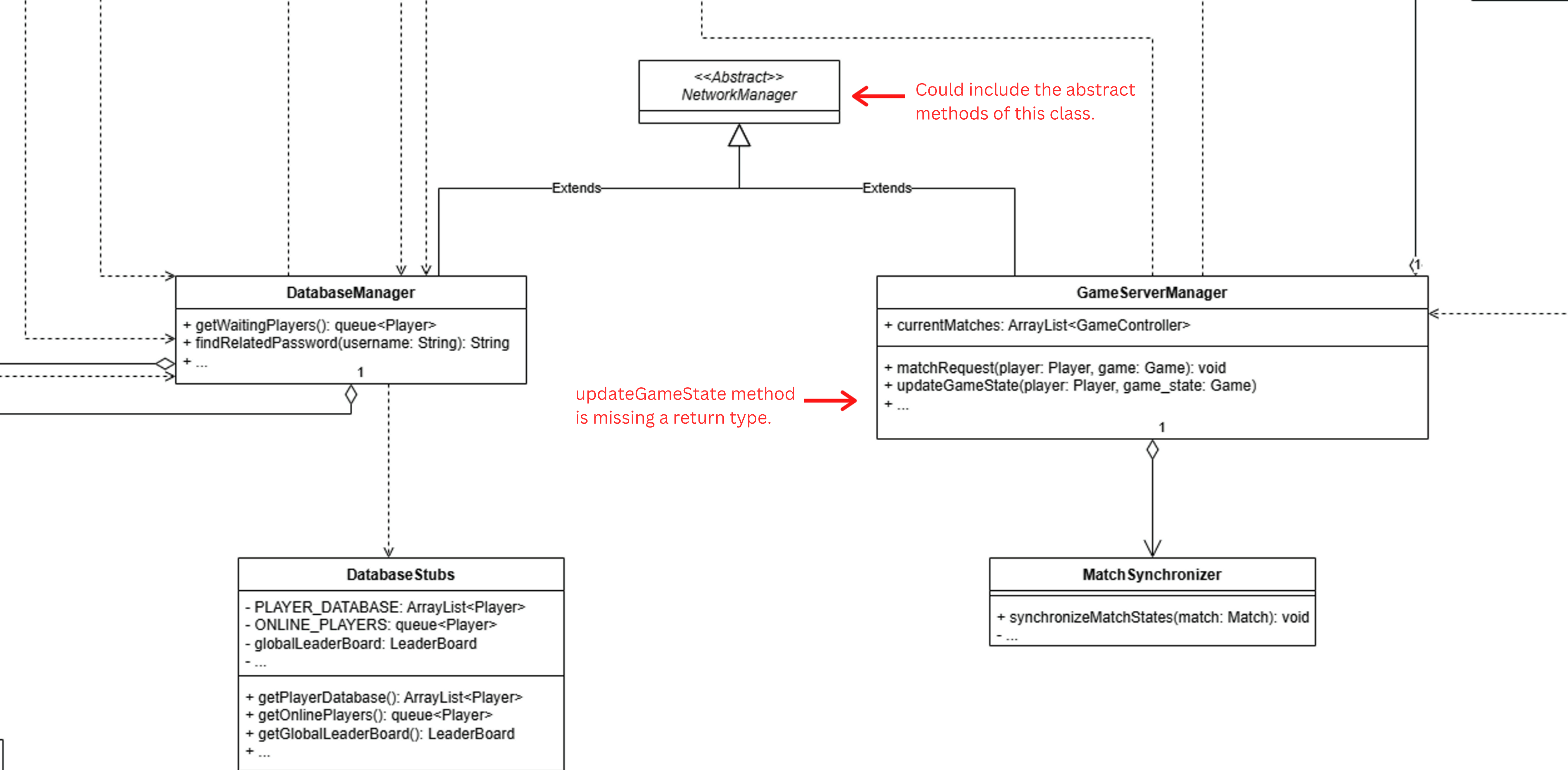
The diagram lacked the case to move a rook( the move rook use case will also extend to castling)

The name of the use case should be check game over as it extends both checkmate(win condition) and stalemate(draw condition)

The label "<<initiates>>" is not standard uml format



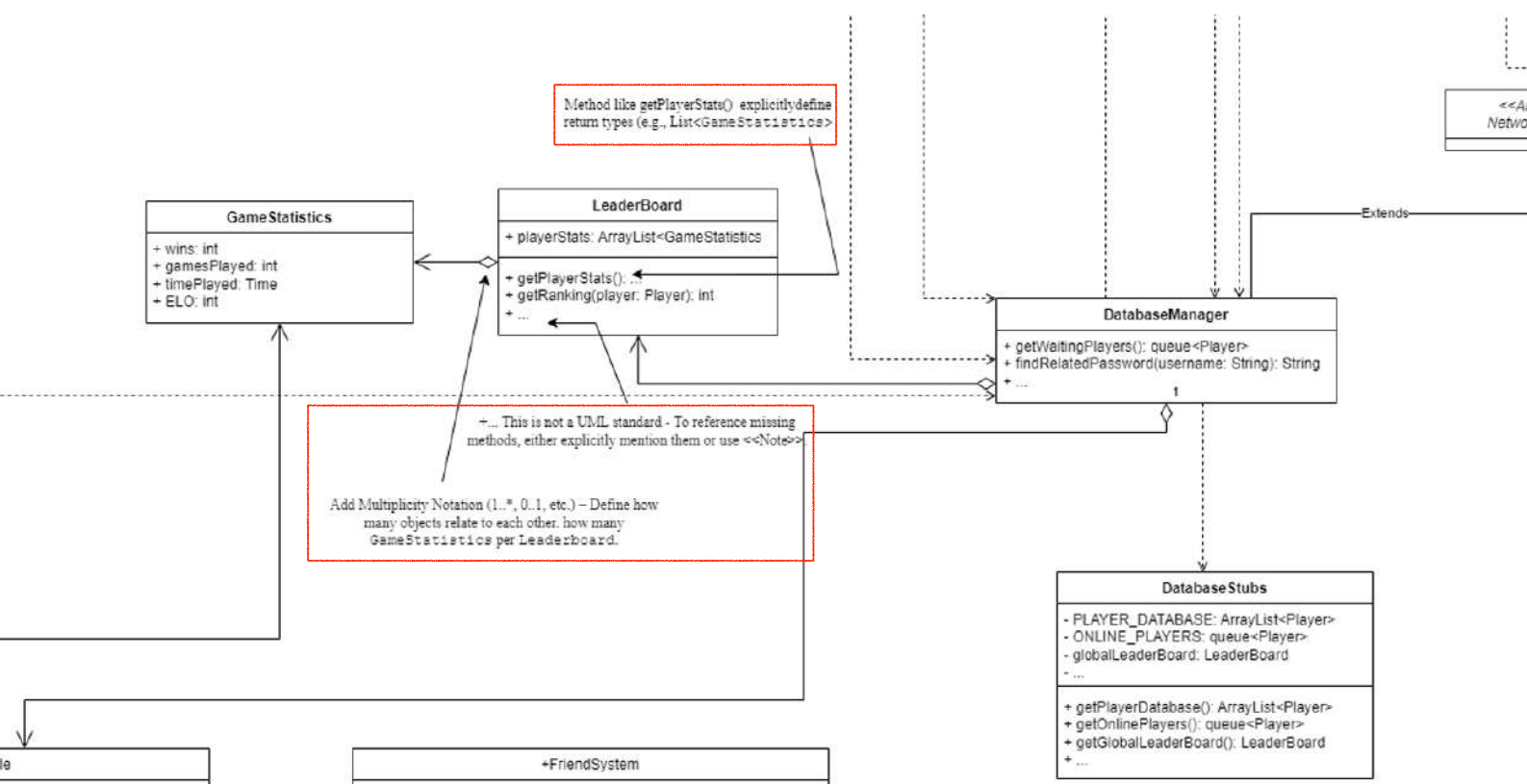
The use case for check draw condition(board full) was not displayed



← Could include the abstract methods of this class.

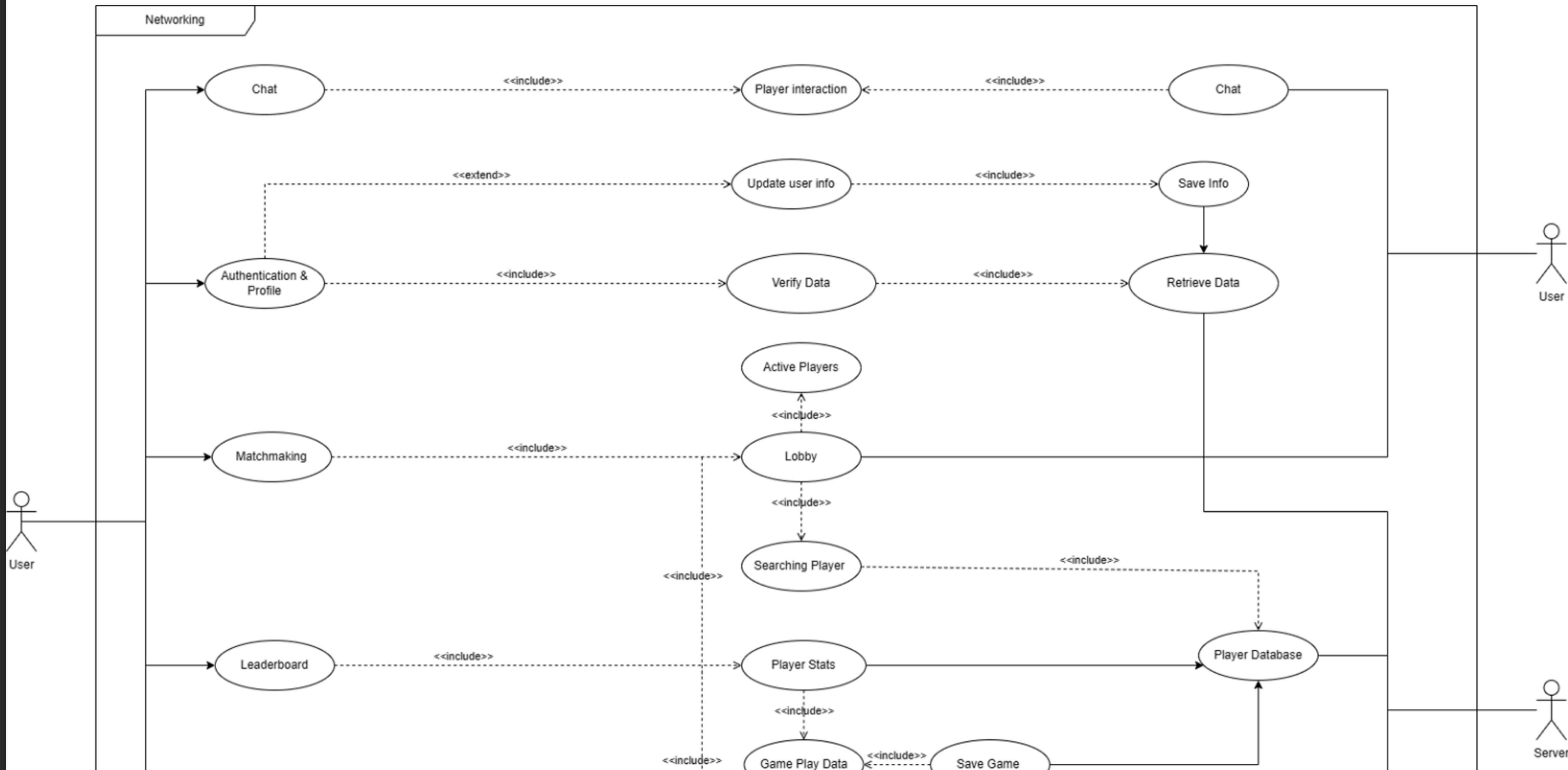
→ updateGameState method is missing a return type.

You could make separate diagrams for different components of the project to improve readability of the diagram.



There are two actors named 'User'. If they are different actors, you could rename them to make it clear. If it is the same actor, you could remove one and create more arrows.

Similarly, there are two use cases named 'Chat'. You could rename it to make it more clear whether they are same or different.



# Leaderboard & Matchmaking Use Cases

## See Personal Stats

Iteration: 1, last modification: March 7, 2025

Adding personal stats to Matchmaking & Leaderboard would result in low cohesion. It would be preferable to have the authentication & profile team handle it.

|                                    |                                                                                                                                                                                                                                                                                                                  |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                 | Passing the player's data to profile for user to view the personal statistics                                                                                                                                                                                                                                    |
| <b>Primary Actor</b>               | Game Statistics Class                                                                                                                                                                                                                                                                                            |
| <b>Goal in Context</b>             | To pass along data from the database to profile and UI to display the person's data                                                                                                                                                                                                                              |
| <b>Preconditions</b>               | <ol style="list-style-type: none"><li>1. The player must be logged in and the user ID exists in the database</li><li>2. There must be a request from the player via the UI</li><li>3. The data must be converted from SQL to Java</li></ol>                                                                      |
| <b>Trigger</b>                     | The person's data will be converted from Queries to Java then profiles which will be displayed on the UI                                                                                                                                                                                                         |
| <b>Scenario</b>                    | <ol style="list-style-type: none"><li>1. The player clicks on the profile button</li><li>2. It will send a request to Game Statistics Class which will get data from the database and converting it to Java</li><li>3. It will send the data to profile</li><li>4. Profile will display the data on UI</li></ol> |
| <b>Exceptions</b>                  | <ol style="list-style-type: none"><li>1. Data not found</li></ol> - Personal data was not found in the database                                                                                                                                                                                                  |
| <b>Priority</b>                    | High. This is a core system requirement                                                                                                                                                                                                                                                                          |
| <b>When available</b>              | Third Iteration                                                                                                                                                                                                                                                                                                  |
| <b>Frequency of Use</b>            | Once a day                                                                                                                                                                                                                                                                                                       |
| <b>Channel to Actor</b>            | Interaction with the user profile menu UI                                                                                                                                                                                                                                                                        |
| <b>Secondary Actors</b>            | Database                                                                                                                                                                                                                                                                                                         |
| <b>Channel to Secondary Actors</b> | Queries to the database                                                                                                                                                                                                                                                                                          |
| <b>Open Issues</b>                 | <ol style="list-style-type: none"><li>1. How to get data from the database team, how will we communicate with the database team</li><li>2. What type of data should we pass to user profile</li></ol>                                                                                                            |

a high priority system requirement would be better moved to second iteration to give time for testing. I would also reconsider the priority rating, as the feature is more important to player experience rather than core system requirement

# Match Making

Iteration: 1, last modification: March 7, 2025

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                              |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                 | To put the player into a queue to find a game by changing the player's status as interested in the database                                                                                                                                                                                                                                                                                            |                                                                                                                                                                              |
| <b>Primary Actor</b>               | Player                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                              |
| <b>Goal in Context</b>             | To put the player in a match's queue and change their status in the database as interested                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                              |
| <b>Preconditions</b>               | <ol style="list-style-type: none"><li>1. The player must be logged in with a valid ID</li><li>2. Internet connection to the database must be made</li><li>3. The player must select a game that they are interested in playing</li></ol>                                                                                                                                                               |                                                                                                                                                                              |
| <b>Trigger</b>                     | Update database and change the player status in a game to be interested                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                              |
| <b>Scenario</b>                    | <ol style="list-style-type: none"><li>1. The player clicks on a game they want to play</li><li>2. The player clicks on the find a match button</li><li>3. That will send an update to the match making database class</li><li>4. That will send a Query to the database</li></ol>                                                                                                                      |                                                                                                                                                                              |
| <b>Exceptions</b>                  | <ol style="list-style-type: none"><li>1. Player ID not found<ul style="list-style-type: none"><li>- The player ID does not match the player ID</li></ul></li><li>2. No game selected<ul style="list-style-type: none"><li>- The player did not select a game</li></ul></li><li>3. Suitable opponent is not found<ul style="list-style-type: none"><li>- No player with similar elo</li></ul></li></ol> |                                                                                                                                                                              |
| <b>Priority</b>                    | High. This is an important function since this would be the easiest way to enter a match                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                              |
| <b>When available</b>              | Third Iteration                                                                                                                                                                                                                                                                                                                                                                                        | The availability for an important feature required to play the games would be better accomodated by implementing in the first iteration. This leaves enough room for testing |
| <b>Frequency of Use</b>            | Several times a day                                                                                                                                                                                                                                                                                                                                                                                    | and integration.                                                                                                                                                             |
| <b>Channel to Actors</b>           | Interaction with game selection menu UI                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                              |
| <b>Secondary Actors</b>            | Database                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                              |
| <b>Channel to Secondary Actors</b> | Queries to the database                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                              |
| <b>Open Issues</b>                 | How will we update and find a suitable match using the database                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                              |

# Challenge A User

Iteration: 1, last modification: March 7, 2025

|                                    |                                                                                                                                                                                                                                                                                                             |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                 | Interaction with a friend's invitation to start a game                                                                                                                                                                                                                                                      |
| <b>Primary Actor</b>               | Player                                                                                                                                                                                                                                                                                                      |
| <b>Goal in Context</b>             | To accept or deny a request for a game invite                                                                                                                                                                                                                                                               |
| <b>Preconditions</b>               | <ol style="list-style-type: none"><li>1. The player must be logged in with a valid ID</li><li>2. A request must be sent to the player</li></ol>                                                                                                                                                             |
| <b>Trigger</b>                     | Either abort the Match creation or create a Match variable that will be passed to game logic server code                                                                                                                                                                                                    |
| <b>Scenario</b>                    | <ol style="list-style-type: none"><li>1. The player sends out an invitation to their friend</li><li>2. The player's friends will either accept or deny the request through the UI</li><li>3. .That will either trigger the creation of Match in the server or Not.</li></ol>                                |
| <b>Exceptions</b>                  | <ol style="list-style-type: none"><li>1. Player ID not found<ul style="list-style-type: none"><li>- Either the player or player's friend ID is not found</li></ul></li><li>2. Match no longer exist<ul style="list-style-type: none"><li>- If the match invites times out</li></ul></li></ol>               |
| <b>Priority</b>                    | Medium. This is an important function but automatically finding a match is more important                                                                                                                                                                                                                   |
| <b>When available</b>              | Third iteration                                                                                                                                                                                                                                                                                             |
| <b>Frequency of Use</b>            | Several times a day                                                                                                                                                                                                                                                                                         |
| <b>Channel to Actors</b>           | Interaction with game selection menu                                                                                                                                                                                                                                                                        |
| <b>Secondary Actors</b>            | Player's friends                                                                                                                                                                                                                                                                                            |
| <b>Channel to Secondary Actors</b> | Via networking to send the invite and receive the answer                                                                                                                                                                                                                                                    |
| <b>Open Issues</b>                 | <ol style="list-style-type: none"><li>1. How would we go about sending the invites over the network?</li><li>2. How long should the invite last before canceling the invite</li><li>3. How would we put both people into a queue so the system knows when to start the game with those two people</li></ol> |

Issue no. 3 would have to be handled by the matchmaking system as part of core functionality. I suggest defining the matchmaking system and its functionality more explicitly, as part of planning stage

# View Personal Leaderboard

Iteration: 3, last modification: March 7, 2025

|                        |                                                                                                                                                                                                                                                                                                          |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>     | Make a list of 50 players that are above and below the players current rank and pass the list to GUI                                                                                                                                                                                                     |
| <b>Primary Actor</b>   | Player                                                                                                                                                                                                                                                                                                   |
| <b>Goal in Context</b> | A player wants to view where they are in relation to player at the same skill level                                                                                                                                                                                                                      |
| <b>Preconditions</b>   | <ol style="list-style-type: none"><li>1. The player is logged in with a valid player's ID</li><li>2. The player navigated to their profile</li></ol>                                                                                                                                                     |
| <b>Trigger</b>         | Make a sorted list of 50 players that are directly ranked above and below them <span>I suggest making the trigger an action or a button to view player's personal leaderboard.</span>                                                                                                                    |
| <b>Scenario</b>        | <ol style="list-style-type: none"><li>1. The player logs in</li><li>2. The player navigated to their profile</li><li>3. The player wants to check their standing with players of similar elo</li></ol>                                                                                                   |
| <b>Exceptions</b>      | <ol style="list-style-type: none"><li>1. Invalid player's ID<ul style="list-style-type: none"><li>- If the player's ID does not match the database</li></ul></li><li>2. Sever Interruption<ul style="list-style-type: none"><li>- If the connection with the database is interrupted</li></ul></li></ol> |

|                                    |                                                                                                                                                                          |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Priority</b>                    | Low. This is a quality-of-life feature that will encourage players, and help them to be more engaging with the game                                                      |
| <b>When available</b>              | Iteration 3 <span>Considering the priority and iterations assigned to the other use cases, it would be preferrable to move the availability to a later iteration.</span> |
| <b>Frequency of Use</b>            | However often the players want to know their rank                                                                                                                        |
| <b>Channel to Actors</b>           | Interaction with game UI                                                                                                                                                 |
| <b>Secondary Actors</b>            | Database                                                                                                                                                                 |
| <b>Channel to Secondary Actors</b> | Database's queries to get player's data                                                                                                                                  |
| <b>Open Issues</b>                 | <ol style="list-style-type: none"><li>1. Have not picked a data type to pass to GUI</li></ol>                                                                            |

# View Friend's List

Iteration: 3, last modification: March 7, 2025

|                        |                                                                                                                                                                                                                                                                                                          |                                                                               |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Description</b>     | Returns a sorted list of all the other players that have been friended and sorted them by elo.                                                                                                                                                                                                           |                                                                               |
| <b>Primary Actor</b>   | Player                                                                                                                                                                                                                                                                                                   |                                                                               |
| <b>Goal in Context</b> | To view the player elo in their friend's list                                                                                                                                                                                                                                                            |                                                                               |
| <b>Preconditions</b>   | <ol style="list-style-type: none"><li>1. The player opens the game</li><li>2. The game is connection to the internet</li><li>3. The player is logged in</li><li>4. The game is connected to the internet</li><li>5. The player navigates to the friend's list</li></ol>                                  |                                                                               |
| <b>Trigger</b>         | Gives an up-to-date list of all their friends sorted by their elo                                                                                                                                                                                                                                        | This description better fits the post-condition, in comparison to the trigger |
| <b>Scenario</b>        | <ol style="list-style-type: none"><li>1. The player logs in</li><li>2. The player navigates to the friend's list</li><li>3. The player wants to check where they stand in elo ranking to their friends</li></ol>                                                                                         |                                                                               |
| <b>Exceptions</b>      | <ol style="list-style-type: none"><li>1. Invalid player's ID<ul style="list-style-type: none"><li>- If the player's ID does not match the database</li></ul></li><li>2. Sever Interruption<ul style="list-style-type: none"><li>- If the connection with the database is interrupted</li></ul></li></ol> |                                                                               |

|                                    |                                                                                                                              |  |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Priority</b>                    | Low. This is a quality-of-life feature that boost the competitiveness of the player and encourage interactions with the game |  |
| <b>When available</b>              | Iteration 3                                                                                                                  |  |
| <b>Frequency of Use</b>            | Dependent on how serious the player is about the game, ranging from after every match to once per day.                       |  |
| <b>Channel to Actors</b>           | Interaction with game UI.                                                                                                    |  |
| <b>Secondary Actors</b>            | None                                                                                                                         |  |
| <b>Channel to Secondary Actors</b> | N/A                                                                                                                          |  |
| <b>Open Issues</b>                 | Coordinating with GUI                                                                                                        |  |

# View Top Players Leaderboard

Iteration: 2, last modification: March 7, 2025

|                         |                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                           |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>      | Provides a list of global player base sorted by elo                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                           |
| <b>Primary Actor</b>    | Player                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                           |
| <b>Goal in Context</b>  | To have a list of the top 500 players sorted by elo                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                           |
| <b>Preconditions</b>    | <ol style="list-style-type: none"><li>1. The player opens the game</li><li>2. The game is connection to the internet</li><li>3. The player logs in</li><li>5. The player navigate to the leaderboard</li></ol>                                                                                           |                                                                                                                                                                                                                           |
| <b>Trigger</b>          | Show the leaderboard with the top 500 players                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                           |
| <b>Scenario</b>         | <ol style="list-style-type: none"><li>1. The player logs in</li><li>2. The player selects the leaderboard button</li><li>3. The game will show a list of the top 500 players that the player can scroll through</li></ol>                                                                                |                                                                                                                                                                                                                           |
| <b>Exceptions</b>       | <ol style="list-style-type: none"><li>1. Invalid player's ID<ul style="list-style-type: none"><li>- If the player's ID does not match the database</li></ul></li><li>2. Sever Interruption<ul style="list-style-type: none"><li>- If the connection with the database is interrupted</li></ul></li></ol> |                                                                                                                                                                                                                           |
| <b>Priority</b>         | Low. This is a quality-of-life feature that can be implemented later                                                                                                                                                                                                                                     |                                                                                                                                                                                                                           |
| <b>When available</b>   | Iteration 3                                                                                                                                                                                                                                                                                              | I suggest pushing back the availability iteration, as high, medium and low priority features are all listed in the third iteration for all features. This would not leave enough time for proper testing and integration. |
| <b>Frequency of Use</b> | Low, once per day                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                           |

|                                    |                           |
|------------------------------------|---------------------------|
| <b>Channel to Actors</b>           | Interaction with game UI. |
| <b>Secondary Actors</b>            | None                      |
| <b>Channel to Secondary Actors</b> | N/A                       |
| <b>Open Issues</b>                 | Coordinating with GUI     |